IN3060/INM460 Computer Vision Coursework report

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Data preprocessing

- Data Preprocessing was done depending on the models and feature extraction techniques used.
- For models using SIFT, images were converted to greyscale with no resizing and normalisation since SIFT operates on intensity information and is scale invariant. Bag of Visual Words (BoVW) used k-means clustering (k=100) to build the vocabulary.
- For HOG, images were resized to 128x128, converted to greyscale and standardised. HOG features were extracted with 9 orientations, 8x8 pixels per cell and 2x2 cells per block.
- For CNN, we explored pretrained models (MobileNetV3, VGG Net, etc); for which images were resized to 224x224, converted to RGB and normalised using ImageNet mean. Initially, data augmentation such as horizontal flip was applied to the training data.

Implemented models

- **Model 1: SIFT + SVM (with SVM Grid Search for hyperparameter tuning)**
  The base model struggled for Classes 0 and 2. Grid search on hyperparameters 'C', 'gamma' and 'kernel' did not perform as good as the best model which was run using augmented data for 0 and 2.
- **Model 2: HOG + SVM (with SVM Grid Search for hyperparameter tuning)**
  Base model struggled again for minority classes. Best model was obtained through Grid Search. SMOTE and data augmentation did not improve performance.
- **Model 3: CNN (VGG Net)**
  Base model was done with Cross-Entropy loss (weights adjusted based on class distribution), ADAM optimiser and learning rate that would decrease based on accuracy. Focal Loss without weight adjustment improved performance along with data augmentation.
  To mitigate overfitting, dropout on fully connected layer and weight decay (L2 norm) were used.
  The best model was obtained after fine tuning, where the final convolutional block was unfrozen.

Quantitative results

	Accuracy	*Precision*	*Recall*	*F1 Score*
Model 1	*0.68*	*0.46*	*0.60*	*0.47*
Model 2	*0.70*	*0.53*	*0.55*	*0.50*
Model 3	*0.93*	*0.76*	*0.80*	*0.77*

Discussion

- The SVM models, set up with class weight as 'balanced' helped improve performance.
- Grid search was done to find the model with the best recall rather than accuracy due to imbalance data.
- SIFT underperformed due to its inability to capture global features unlike HOG or Haar wavelets.
- HOG was expected to perform a lot better, but the model struggled to generalize.
- MobileNetV3 was our initial choice as a backbone for its balance accuracy and speed; but we achieved suboptimal performance even after implementing Focal Loss and Weighted Random Sampler.
- Focal loss is a version of cross-entropy that down-weights well-classified labels and focus training on difficult cases by using a modulating factor. It is ideal for imbalanced datasets.
- We opted for VGG Net for its deep yet simple architecture with small convolutional filters that we expected would be excellent where fine details and global context are both important.
- All three models generalised quite well on the test data with VGG Net achieving 93% accuracy. The model using SIFT struggled to predict minority classes. The same is observed to less extent for HOG.
- When applied on a video, our best model predicted 'No Mask' and 'Mask' very well while 'Improper Mask' was detected for a few frames only.